



1 COMPUTER VISION & LOCALIZATION

High-Speed Vision & Goal Detection

Our vision pipeline utilizes the OpenMV H7 Plus camera to process real-time field data at 45+ FPS. To maximize performance, we implemented hardware-level sensor flipping, allowing full orientation correction without CPU overhead. The system detects goals using optimized color thresholds and geometric filtering, enabling accurate bearing estimation even during rapid movement.

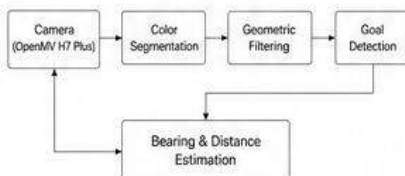
Camera-Guided Goal Alignment

Once the ball is secured, the robot transitions to a vision-guided targeting mode. The system continuously analyzes the goal position and applies proportional steering corrections. Distance is estimated using the vertical position of the goal in the frame, allowing dynamic adjustment of shooting force for improved accuracy.

Continuous Vector Pursuit Algorithm

To eliminate instability caused by discrete logic states, we implemented a continuous vector-based pursuit algorithm. The robot calculates a full 360° movement vector relative to the ball, enabling smooth orbital motion. This approach maintains optimal positioning while preserving speed and efficiency.

VISION PIPELINE



The pipeline outputs real-time bearing and distance information for precise robot control and target alignment.

2 SENSOR FUSION & IR DETECTION

IR Detection System

The robot uses 24 TSSP4038 infrared sensors, each providing an independent pulse signal. The system determines the ball direction by selecting the sensor with the strongest signal and multiplying its index by 16°, achieving full 360° coverage.

Distance estimation is calculated using the sum of all sensor readings combined with a calibrated formula, providing a reliable approximation of ball proximity.

Multiplexing Architecture

Due to the large number of sensors, two CD74HC4067 multiplexers are used to reduce pin usage. This configuration compresses 24 inputs into a small number of digital and analog lines.

The custom PCB ensures stable communication between components, enabling reliable operation of the full sensor array.

High-Speed Signal Processing

Using the Teensy 4.1, the system samples IR data at approximately 40 Hz to match fast gameplay conditions. The multiplexers rapidly switch channels, allowing efficient sequential reading of all sensors with minimal delay.



3 ELECTRONICS & CUSTOM PCB

System Controller

The robot is built around the Teensy 4.1 microcontroller, which serves as the central processing unit. It was selected for its high processing speed, reliability, and ability to handle real-time computation.

Compared to standard boards such as Arduino Uno, it provides significantly more computational power, memory, and faster communication interfaces, enabling simultaneous sensor processing and motor control.

Custom PCB Design

A custom PCB was developed to integrate sensing, processing, and power systems into a single platform.

The board includes:

- Two CD74HC4067 multiplexers for sensor expansion
- Support for 24 TSSP4038 IR sensors
- A Bosch BNO055 IMU for orientation tracking

The IMU ensures accurate heading information, allowing the robot to maintain directional awareness even during movement or collisions.

Power Management

Power regulation is handled using two voltage regulators:

- D36V50F3
- D36V50F5

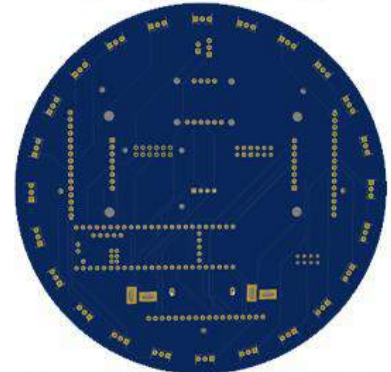
These provide stable voltage to all subsystems. Proper regulation prevents noise, instability, and unexpected resets during operation.

Motor Control

Motor drivers are used to control the robot's motors. The Teensy sends PWM signals to the drivers, which regulate power delivery.

This enables precise control of speed and direction, allowing smooth motion, rapid acceleration, and accurate maneuvering.

CUSTOM PCB LAYOUT (TOP VIEW)



4 MECHANICAL ENGINEERING

Omni-Directional Chassis Design

The robot is built on a 4-wheel Omni-X configuration, enabling full holonomic movement. This allows independent control of translation and rotation for maximum maneuverability.

The chassis combines lightweight aluminum with high-strength 3D-printed components, achieving a strong and efficient structure.

Center of Gravity & Stability

Key components such as the battery and motor drivers are positioned low in the chassis to reduce the center of gravity. This improves stability and prevents tipping during high-speed maneuvers.

Precision motor mounts reduce vibration and increase efficiency of power transfer.

5 AUTONOMOUS STRATEGY

Hierarchical Decision Framework

The robot operates using a hierarchical control system based on sensor confidence from both IR and vision systems.

- **Search & Circumvent:** The robot orbits the ball to reach an optimal shooting position without unintended contact.
- **Precision Striking:** Once aligned, the robot switches to vision-based targeting for accurate shooting.

Adaptive Field Awareness

A fail-safe system continuously monitors IMU and boundary sensors. If a collision or out-of-bounds condition is detected, the robot performs an immediate corrective maneuver.

This ensures continuous gameplay while maintaining ball tracking and performance.

TECHNICAL SPECIFICATIONS

Processor	Teensy 4.1
Vision Sensor	OpenMV H7 Plus
IR Array	24x TSSP4038 Sensors (360° Coverage)
IMU Sensor	BNO055
Drive System	4x High-Torque DC Motors + Omni-Wheels
Power Input	11.1V 3S LiPo Battery

